DA3 · DEEP LEARNING & GENAI · FINAL ASSIGNMENT

LexiSyncAI

An evidence-grounded legal document analyst: answers that cite page and clause, or abstain

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github.com/hamzakazi511-eng/LexiSyncAI

The problem: legal answers must be verifiable

Every claim cited

Document, page and clause for each statement, highlighted on the page

Multi-hop

Combine clauses across pages and documents, and resolve amendments

Abstain

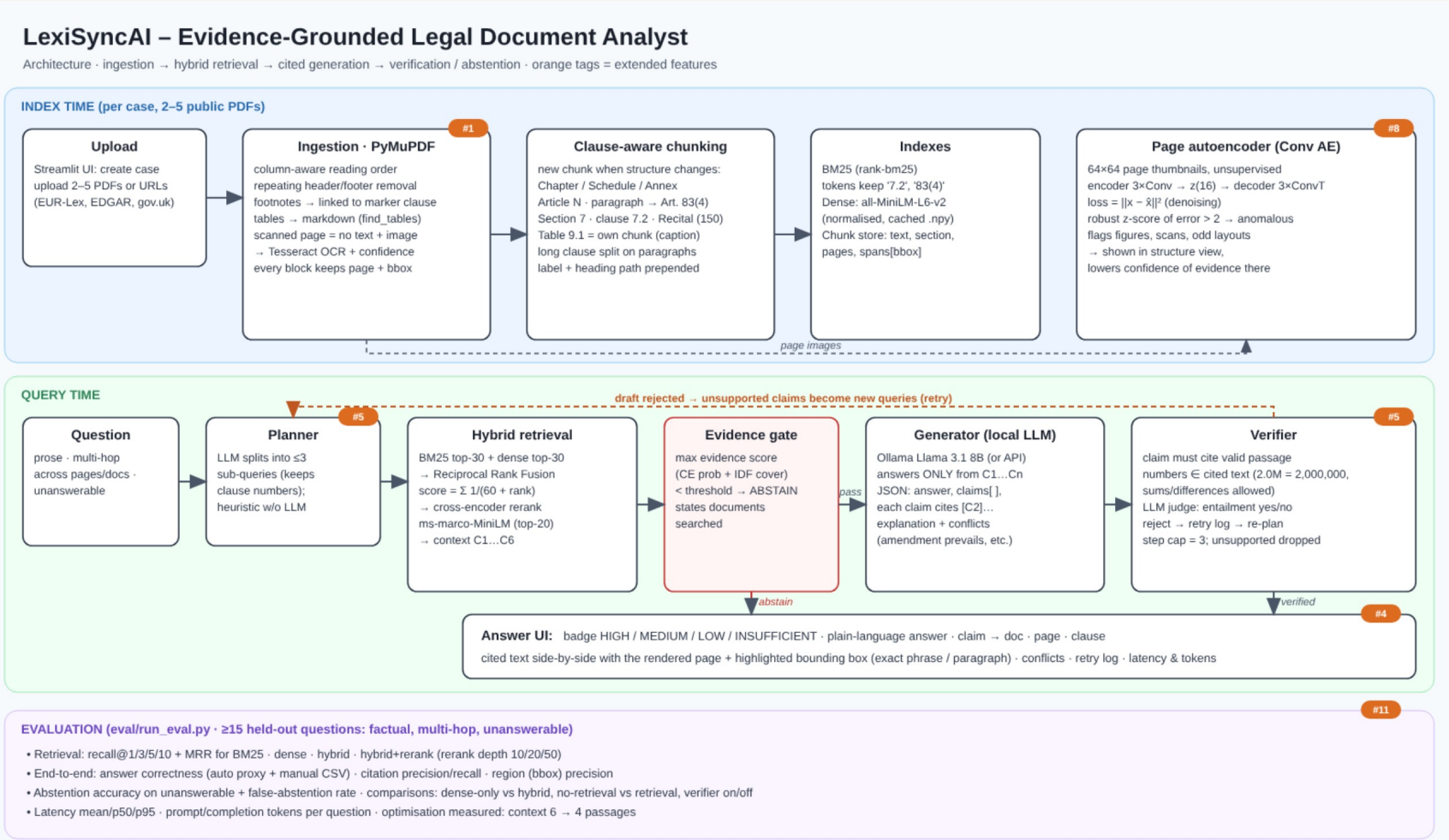
Say “insufficient evidence” instead of guessing, and name the documents searched

Example question

“Does the liability cap in Section 7 exceed the total indemnity obligations in Section 9?”

No: cap EUR 2,000,000 (§7.2, p. 4) < indemnities EUR 3,500,000 (Table 9.1, p. 4)

Architecture



Ingestion and clause-aware chunking

Reading order: column-aware, left column before right

Headers and footers: repeating margin text and page numbers removed

Footnotes: attached to the clause that carries the marker

Tables: extracted cell by cell as markdown, one chunk each

Scanned pages: no text + image → Tesseract OCR (confidence 0.95)

New chunk when the structure changes

Chapter · Schedule · Annex

Article 83 → paragraph “4.” → **Article 83(4)**

Section 7 → clause **7.2** · Recital (150)

Every chunk keeps **page + bounding box** per paragraph

Not fixed-size windows: a citation can always name page and clause

Hybrid retrieval with cross-encoder reranking

BM25 · top 30

Exact identifiers: “7.2”, “83(4)”, “EUR 2,000,000”

Dense MiniLM · top 30

Meaning: “cap” ≈ “shall not exceed”

→

RRF fusion

score = Σ 1 / (60 + rank)

Rank-based, so BM25 and cosine scores mix safely

→

Cross-encoder rerank

ms-marco-MiniLM on the top 20 → context C1…C6

Query and chunk attended jointly

RAG lecture: candidate retrieval maximises **recall**, reranking maximises **precision**

Grounded generation, verification and abstention

1 · Plan

Split the question into up to 3 sub-queries, one per clause

2 · Draft

Local Llama 3.1 8B answers only from C1…Cn as JSON claims, each citing ids

3 · Verify

Numbers must be in the cited text · LLM judge · citation repair · retry up to 3

4 · Answer

Badge HIGH / MEDIUM / LOW, conflicts explained, page shown with highlight

Abstain (INSUFFICIENT EVIDENCE) when

the best evidence score is below the threshold · the question names a document that is not in the case · no claim survives verification

Extended features: 5 of 11

#1 OCR for scanned pages

Image-only pages detected, Tesseract with word boxes; confidence 0.95 on the signature page

#4 Bounding-box citations

Exact amount or phrase boxed on the page image; region precision 0.83

#5 Planner + verifier loop

Rejected drafts logged and retried; +7 points correctness, citation precision 1.00

#8 Convolutional autoencoder

Page-anomaly detector on 64×64 thumbnails; chart page z = 19.8, text pages below 1.1

#11 Latency and cost report

Seconds and tokens per question; rerank depth 10 instead of 50 is 8× faster

Retrieval: page-level recall@k per configuration

Configuration	recall@1	recall@3	recall@5	MRR	latency
BM25	0.800	0.933	1.000	0.967	1 ms
Dense (MiniLM)	0.867	0.967	1.000	1.000	13 ms
Hybrid (RRF)	0.800	1.000	1.000	0.967	9 ms
Hybrid + cross-encoder (depth 20)	0.867	0.933	1.000	1.000	671 ms
Hybrid + cross-encoder (depth 10)	0.867	0.933	1.000	1.000	139 ms
Hybrid + cross-encoder (depth 50)	0.867	0.933	1.000	1.000	1121 ms

All evidence is in the top 5 for every configuration; reranking gives the best top-1. Depth 10 keeps the same recall at 8× less latency.

End to end: retrieval and hybrid search matter

0.80

Full system: answers correct

0.53

Dense-only retrieval (multi-hop 0.00 vs 0.60)

0.13

LLM alone, no retrieval: never abstains

1.00

Citation precision: every cited page is a gold page

Manual grading **15 / 19** · factual 9/10 · multi-hop 3/5 · unanswerable 3/4 abstained · region precision 0.83 · no false abstentions

Cost of each component (local model, 0 USD)

Configuration	correct	p50 latency	prompt tokens / q	context tokens / q
Full system	0.800	12.0 s	1306	500
Without verifier	0.733	8.5 s	923	494
Context 4 instead of 6 passages	0.667	11.0 s	1264	395
No retrieval	0.133	4.3 s	96	0

Verifier

+3.5 s and ~380 tokens per question buy +7 points and perfect citation precision

Smaller context

−21 % context tokens and 4/4 abstention, but multi-hop falls from 0.60 to 0.40

Error analysis and next steps

Miss	What happened	Next step
F10 footnote	Footnote retrieved, base rate not named	Quote footnote text in the answer
M02 fee increase	New fee given, increase not computed	Arithmetic tool for the agent
M04 two-part question	One of two parts answered	Check every sub-query is answered
U03 unknown role	Answered instead of abstaining	Entity and role check before drafting

Limits: small synthetic bundle (48 chunks) · 12 s per answer on a CPU · tables need ruling lines · thresholds tuned on one bundle

Live demo

1 · Upload and index

Unseen public PDF → structure view: clauses, tables, flagged pages

2 · Three questions

Prose · two-page multi-hop · unanswerable

3 · Evidence

Highlighted page, verifier log, confidence badge

Reproduce: make run · docker compose up · pinned dependencies · seed 42

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